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NAO ROBOT EXERCISE SYSTEM

Project Description and Goals

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Bratislava • 2026/2027

Contents

1 Project Description **2**

2 Overall Goal **2**

3 Project Objectives **2**

3.1 Video-based choreography 2

3.2 Robot stability and fall prevention 2

3.3 Actuator and posture constraints 3

3.4 Service architecture and maintainability 3

3.5 Interaction and sensing reliability 3

4 Expected Outcomes **3**

1 Project Description

The NAO Robot Exercise System is an existing platform for guided physical exercise, especially for older adults. NAO demonstrates movements, visual pose processing observes the participant, and the system provides feedback. A trainer-facing interface prepares and manages exercises. The components communicate as services.

The NAOMI team will extend this platform rather than build a new one. Work focuses on video-based choreography, movement stability, thermal limits, and service reliability. The NAOqi interface requires Python 2 alongside newer components. Because robot access is intermittent, changes must be testable without hardware and checked on the robot when available.

The team will document implemented improvements, test evidence, and remaining limits.

2 Overall Goal

Improve the existing NAO exercise system so that a trainer can prepare and run exercise sessions more reliably, while robot movement remains within experimentally established safety and hardware limits.

3 Project Objectives

3.1 Video-based choreography

Support the preparation of a choreography from multiple videos. A trainer should be able to select videos, trim relevant segments, arrange them into one exercise sequence, and preview the resulting movements before they are sent to the robot. The delivered workflow and its limitations will be demonstrated with representative source videos.

3.2 Robot stability and fall prevention

Investigate when NAO loses balance during exercise. Test movement amplitudes and transitions when the robot is available, then use the findings to constrain generated movements. Document repeatable scenarios, failures, and observed safe ranges.

3.3 Actuator and posture constraints

Study the effect of sustained postures and repeated movements on actuator temperature. Identify exercise patterns that may create excessive thermal load and document practical limits or safeguards for choreography design. Findings will be based on recorded robot measurements where feasible.

3.4 Service architecture and maintainability

Review how the system's services exchange data and commands. Reduce avoidable dependencies, make service boundaries and failures clearer, and keep the NAOqi-dependent Python 2 component isolated from services that use newer Python environments. Document the revised interfaces and verify affected service interactions with repeatable tests.

3.5 Interaction and sensing reliability

Evaluate the limitations of pose and emotion recognition in the current workflow, including the system's dependence on the Intel RealSense depth camera. Improve handling of unreliable or unavailable sensor input where feasible, and document which interaction scenarios are supported and which remain limited.

4 Expected Outcomes

The project will deliver an enhanced version of the existing exercise system, including the implemented choreography and service improvements. It will also produce documented findings on robot balance, movement limits, actuator temperature, and sensing reliability.

The team will record software test results and, when NAO access permits, each hardware scenario and outcome. Final documentation will describe reproducible workflows, verified behaviour, and remaining limits.